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Children with autism spectrum disorder and social skills groups at school: a randomized trial comparing intervention approach and peer composition

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Background: Peer relationships improve for children with autism spectrum disorder (ASD) in clinic-based social skills groups but rarely generalize to real world contexts. This study compares child outcomes of two social skills interventions conducted in schools with children in Kindergarten through fifth grade. **Method:** Children with ASD were randomized to one of two interventions that varied on group composition (mixed typical and ASD vs. all ASD or social difficulties) and intervention approach (didactic SKILLS based vs. activity-based ENGAGE groups). Interventions were implemented at school for 8 weeks (16 sessions) with an 8-week follow-up. Innovative measures of peer nomination and playground peer engagement, as well as teacher reports of child behavior problems and teacher–child relationship were analyzed for 137 children with ASD across four sites. **Results:** On the primary outcome of social network connections from the peer nomination measure, there was no main effect of treatment, but there were moderator effects. Children with low teacher–child closeness or high conflict improved more in their social connections if they received the SKILLS intervention, whereas children with higher teacher–child closeness improved more if they received the ENGAGE intervention. Only two secondary outcome measures yielded significant effects of treatment. Children in the SKILLS groups increased peer engagement and decreased isolation during recess. Child behavior problems and teacher–child closeness moderated peer engagement such that children with higher behavior problems and lower closeness benefitted more from SKILLS groups. **Conclusions:** These findings suggest that social skills groups conducted at school can affect both peer engagement during recess as well as peer acceptability. Child characteristics and teacher–child relationship prior to intervention yield important information on who might benefit from a specific social skills intervention. **Keywords:** Social skills groups; autism spectrum disorders; inclusion; peer relationships; teacher child relationship; social networks.

Introduction

Establishing positive peer relationships may be the most complex challenge facing children with autism spectrum disorder (ASD). Children with ASD report they have fewer friends at school, less friendship reciprocity, and smaller social networks compared to their classmates (Bauminger, Shulman, & Agam, 2003; Kasari, Locke, Gulsrud, & Rotheram-Fuller, 2011). Despite strengths in academic abilities for many children with ASD, their difficulties in social interactions can lead to poor school adjustment.

Friendships among neurotypical children

Drawing from research on neurotypical children, the majority of school-aged children average five friends, at least one of whom may be a best friend (Hallinan, 1980). These friendships are marked by closeness, reciprocity, and companionship. Moreover, friendships do not exist in isolation. They are embedded within a larger group of social networks, highlighting the importance of social relationships within the school setting.

There are two theoretical explanations regarding why certain children connect to each other. Propinquity refers to being in close proximity; it is essential to friendship development, but not sufficient. Homophily suggests that children will connect to each other along common characteristics, such as gender, ethnicity/cultural background, and common interests. Children in the same classroom who share common interests and characteristics demonstrate both propinquity and homophily. These characteristics are found among children who are friends versus nonfriends (Berndt, 2002).

Friendships and social networks among children with ASD

For children with ASD, both homophily and propinquity have similar push and pull on their relationships. Being in close proximity, having similar characteristics and sharing common interests are

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key to developing friendships for all children, but a challenge in inclusion classrooms is that there is usually only a single child with ASD in the class. Because the majority of contact is with neurotypical children, it is unclear how well propinquity and homophily needs are met for children with ASD. Indeed, to date, studies of inclusion have reported limited improvements in friendship development for children with ASD (Kasari et al., 2011).

Interventions for social skills

To address social difficulties at school, many families obtain social tutoring for their child in the form of clinic-based, social skills groups. The group-based curriculum often follows a set of linked topics including how to start and maintain a conversation, handle teasing, and choose appropriate friends (Laugeson et al., 2012). These types of interventions have shown efficacy on a range of outcomes, but rarely are gains in the context of the group tested for generalization to the child's everyday environment of school (Bellini, Peters, Benner, & Hope, 2007; Reichow, Steiner, & Volkmar, 2013). This is somewhat surprising as often the reason for tutoring is to influence skills that will be used at school and in the community.

There are likely at least two reasons why social groups conducted away from school have limited effects within school. One is that the group composition of most clinical groups consists of all individuals with ASD from different classes, schools, and communities. Therefore, developing a friend within the group may not be sustainable outside the group due to lack of propinquity. Second, the skills addressed may need to be individualized within the context of the group in order to generalize. Skills addressed within the group can be a mismatch to the child's needs within the school environment. For example, a child may have limited knowledge of the games played at school that keep him from participating, but these specifics are not known or included in the group curriculum.

Group social skills interventions carried out in school may have better results for children's friendships and social networks because children in the groups come from the same school, and the curriculum can address concerns that are evident in the school setting. Two recent effectiveness trials provide some evidence for improvements in social skills when interventions are conducted in schools, although neither study included social network or friendship measures (Beaumont, Rotolone, & Sofronoff, 2015; Kenworthy et al., 2014). Thus, the goal of this study was to compare two different social skills interventions that varied group composition and teaching approach of social skills content with effects measured in terms of real world child social skills.

Group composition. Based on our previous work in schools, peer-mediated interventions were significantly more effective for social network connections in the class than were adult-mediated, one-on-one interactions (Kasari, Rotheram-Fuller, Locke, & Gulsrud, 2012). In this study, the typical peers were taught independent of the children with ASD. An open question is whether working in small groups with typical peers and the child with ASD from their class versus a small group of all children with ASD from different classes would be more effective in improving peer acceptance and engagement? In this study we constructed these groups in an effort to determine if social outcomes might vary depending on group composition.

Teaching approach. Clinic-based social skills groups are often didactic in nature with some modeling and practice within the group. One social skills group approach (SKILLS) in this study was modeled after a didactic, clinic-based social skills group. In contrast, other researchers have studied social engagement among peers by capitalizing on shared interests between children at school. In this situation, shared interest activities are provided to give children experiences together, and specific social skills are taught naturalistically within this context (Koegel, Vernon, Koegel, Koegel, Paullin, 2012; Wolfberg, DeWitt, Young, & Nguyen, 2015). This approach is more often used with typical peer models who are chosen because they have good social skills. Our second group (ENGAGE) was based on shared activities between typical children and the child with ASD from the same classroom.

Teacher perceptions. Teachers are critical reporters of child behavior in school, but few studies have linked teacher report to child performance. A previous study found that teachers reported changes in social skills for children with ASD who received different social skills interventions at school (Kasari et al., 2012), and it is likely that teachers might have important insights on child behaviors that predict how well children will do in different interventions. In particular, child behavior problems and the teacher's attachment to the child may predict child performance (Robertson, Chamberlain, & Kasari, 2003).

In this study, we contrasted groups composed of children with ASD from different classes, or mixed groups of typical children and children with ASD from the same classes. Teaching approach was also contrasted between didactic, direct instruction on social skills and shared activity based, naturalistic teaching of social skills. The primary outcome was social network connections based on peer report. Secondary measures included peer engagement on the playground, number of peers nominated as friends by the child with ASD and number of peers nominating the child with ASD as a friend. Given our previous study, we expected the ENGAGE group to have a greater effect on both the primary and secondary outcomes. Further we explored a number

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of potential moderators of outcome including child behavior problems and teacher–child relationship.

Methods

Participants

Study participants included school-aged children with ASD recruited across four sites (Los Angeles, Baltimore, Seattle, Ann Arbor) over the course of 2 years. The trial was registered (clinical.trials.gov NCT01011764) and approved by each site's Institutional Review Board. Families gave written consent, and children assented.

A total of 150 children were screened for participation in the study. Children included in this study had a diagnosis of autism (confirmed by the Autism Diagnostic Observation Schedule (ADOS) and Social Communication Questionnaire (SCQ), an IQ greater than or equal to 65 (confirmed by the abbreviated Stanford Binet-5 (SB-5), between the ages of 6 and 11 years, in elementary school (grades 1–5), with more than half from ethnic/racial minority backgrounds and educated in a general education classroom for a minimum of 80% of the school day. Because two children did not meet criteria for the study, 148 children from 120 classrooms were randomized to the ENGAGE or SKILLS condition (Figure 1).

Eighty-two participants were randomized to the ENGAGE condition and 66 participants received the SKILLS condition. The group size differences are the result of the randomization procedure, which was used to prevent contamination of the intervention (see *Randomization* below). Eleven participants (2 ENGAGE, 9 SKILLS) dropped out of the study after being randomized. The reasons these participants dropped out are attributed to: their schools or classroom teachers were unwilling to participate; they were no longer interested in being in the study, their teachers changed in the middle of the school year, or they moved away. Demographic data from the 137 study participants are presented in Table 1.

Randomization

The children recruited in the first year were randomized separately from the children recruited in the second year because only two sites recruited participants in the first year (Los Angeles, Ann Arbor). The randomization was done separately within site and stratified by grades (Grades 1–2; grades 3–5, and the procedure was designed to control for classroom and treatment group differences). Ninety-three classrooms had one target child with ASD and 27 classrooms had two or three target children with ASD. The SKILLS intervention required that the child with ASD participate in intervention groups with students outside of their classroom. The ENGAGE condition paired targets with ASD with other students from their own class. Therefore, to prevent contamination to the SKILLS intervention, we randomized by classroom and stratified on how many target children were in the class. Classes with only one child were randomized 50–50 to either the SKILLS or the ENGAGE condition. The classes with multiple target children were randomized so that either zero or one child got the SKILLS condition and all other children were randomized to the ENGAGE condition. An independent data management core assigned randomization.

Procedure and intervention

Both intervention conditions occurred at school during morning or lunchtime recess for a total of 16 sessions (2 times per week for 8 weeks), and each session was 30–45 min long. Assessments were collected prior to treatment, at post-treatment, and 8 weeks following the completion of the intervention.

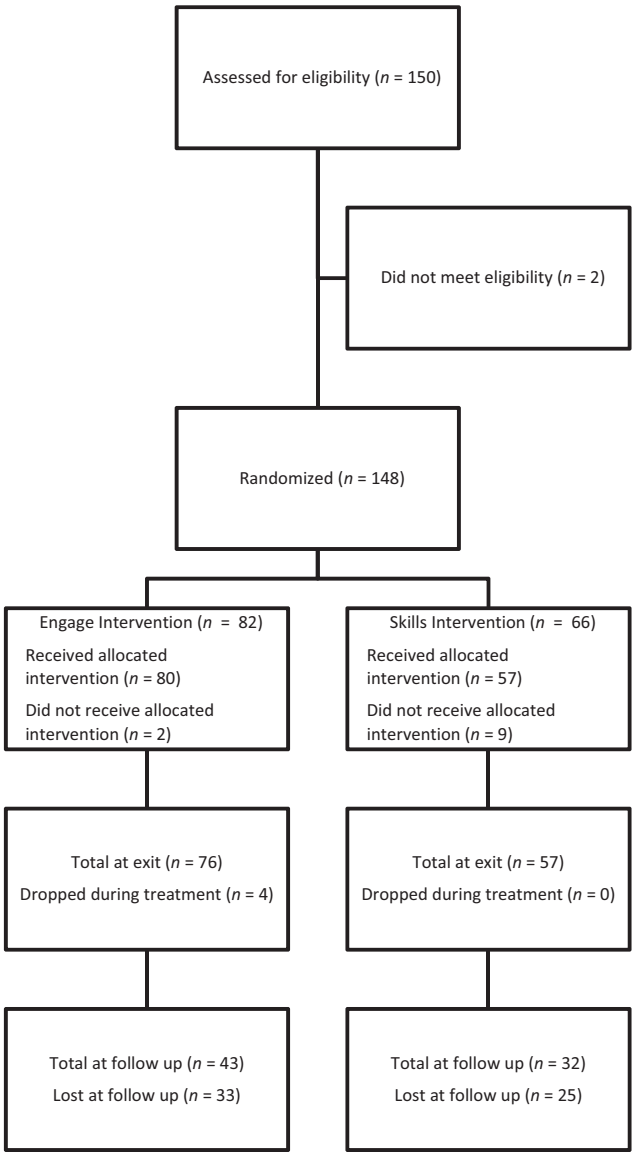


Figure 1 Consort chart

SKILLS. The SKILLS intervention targeted a specific set of social skills. The content delivered included lessons as follows: Being a Social Detective; Greetings and Goodbyes; Body Talk (Nonverbals); Humor; Conversation; Dealing with Teasing; Perspective Taking; Dealing with Emotions; and Friendship Tips. Children were given weekly homework assignments to reinforce the topics discussed in the group sessions.

Each session began with children checking in for the day and reviewing the lesson and homework from the week before. Next, they participated in an interactive lesson during which the topic of the day was introduced and an activity was completed to practice that skill. In each session, children had time for free play with the homework assignment presented at the closing of the session. A treasure box with rewards was distributed every other session (the last session of the week). At every session, group leaders facilitated group rapport, praised skills as they occurred, emphasized including others, and used punch cards for children to earn weekly rewards from the Treasure Box.

For example, during one session, the children learn about keeping a conversation going. Using a didactic format, the group leader introduces conversations by suggesting they are like a bridge—connecting two people together like a bridge connects two pieces of land. One strategy for staying on the bridge is to use a 50/50 rule (reciprocity). Next, the group

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Table 1 Demographic characteristics of children with autism spectrum disorder by treatment groups

Demographics	Engage	Skills	TX *p*-value
Male: *n* (%)	61 (76.25%)	48 (84.21%)	.611
Age: Mean (*SD*)	8.13 (1.55)	8.23 (1.63)	.802
Race: *n* (%)			
African American	4 (5.00%)	6 (11.32%)	.535
Caucasian	39 (48.75%)	23 (40.35%)	
Hispanic	12 (15.00%)	8 (14.04%)	
Asian	14 (17.5%)	14 (24.56%)	
Other	5 (6.25%)	2 (3.51%)	
Did not disclose	6 (7.5%)	4 (7.02%)	
Language at Home: *n* (%)			
English	59 (73.75%)	43 (75.44%)	.999
Spanish	5 (6.25%)	4 (7.02%)	
Other	8 (10.00%)	5 (8.77%)	
Did not disclose	8 (10.00%)	5 (8.77%)	
Stanford Binet 5: Mean (*SD*)			
Nonverbal	9.73 (3.37)	10.47 (3.33)	.211
Verbal	7.95 (3.44)	7.95 (3.99)	.995
IQ	91.98 (16.37)	92.93 (16.12)	.737
ADOS Severity: Mean (*SD*)	7.76 (2.02)	7.18 (2.09)	.103
SSIS: Mean (*SD*)			
Problematic Behavior	112.63 (13.54)	111.15 (12.80)	.555
STRS: Mean (*SD*)			
Closeness	37.13 (7.91)	37.76 (6.00)	.612
Dependency	10.04 (3.11)	9.35 (2.64)	.185
Conflict	24.58 (9.67)	22.04 (7.30)	.097
Site: *n* (%)			
Los Angeles	28 (35.00%)	23 (40.35%)	.681
Baltimore	26 (32.50%)	18 (31.58%)	
Washington	14 (17.50%)	6 (10.53%)	
Michigan	12 (15.00%)	10 (17.54%)	
Entry Outcome Measures: Mean (*SD*)			
POPE Engagement	42.02 (31.07)	37.61 (28.99)	.416
POPE Solitary	31.10 (30.67)	41.44 (29.96)	.026
Out-degree	3.47 (2.65)	3.47 (3.47)	.947
In-degree	1.79 (1.77)	1.47 (1.84)	.110
Social Network Ratio	0.28 (0.25)	0.3 (0.29)	.938

leader facilitated a guided practice activity, where the children and therapist use role play to demonstrate staying on the bridge and not falling into the river below. At the end of the session, each child was given homework (a worksheet), to record his or her use of conversation bridges. The Seattle Children's Hospital Research Institute staff adapted the treatment model for this project.

ENGAGE. The ENGAGE intervention included children with ASD as well as their typically developing peers. There were a greater number of typical peers included to model social behaviors and to foster friendships, typically 2–3 classmates to each child with ASD. The typically developing peers were selected based on results from the friendship survey and teacher nominations, so that they were viewed as positive role models.

The social group targeted peer engagement and acceptance, and used shared interests of the group to provide the context for interactions. Each ENGAGE group meeting consisted of collectively establishing a daily schedule in order to encourage cohesiveness among group members. Activities included conversational exercises, structured games, free play, improvised storytelling, and music. Peers were encouraged to take leadership of their own groups with supervision from adults as needed.

An example of an ENGAGE group begins with the group leader asking children what games they want to play in that session. As a group, the children created a list of activities. First, the students would play a word game for example telephone, building a story. Next, children would play one or two more active games (head's up 7-up, freeze dance), Then, the children would move from the classroom to the playground and continue to play recess games (What time is it Mr. Wolf, 4 square, jump rope). Throughout the session, the group leader facilitated play as needed, but faded out of activities as soon as the children were able to play independently. The UCLA staff developed the intervention manual (Kretzmann, 2008).

Interventionist and treatment fidelity. Interventionists were graduate student researchers and research assistants who were trained in either the SKILLS or the ENGAGE intervention. To maintain the integrity of the interventions, a fidelity checklist captured the core components of the interventions. Interventionists were trained and considered to be at fidelity with a criterion of $\alpha > .80$. Every session was videotaped. The primary creators of each intervention randomly coded 20% of the videotaped sessions for fidelity. The interventionists were consistently above 80% fidelity (SKILLS: $\mu = 86.00$ (1.60); ENGAGE: $\mu = 88.54$ (4.18).

Eligibility measures

The ADOS (Lord, Rutter, DiLavore, & Risi, 2002) was used to assess eligibility into the study. The ADOS is a semistructured, standardized assessment of communication, social interaction and play, and imaginative use of materials. The ADOS-2 Calibrated Severity Metric (CSS), a standardized score that controls for age and language level when describing ASD symptom severity was obtained.

The Social Communication Questionnaire (*SCQ*; Rutter, Bailey, & Lord, 2003) is an assessment used to screen for the presence of ASD symptoms. Parents complete a 40-question survey about their child's developmental history, resulting in a total score that reflects ASD characteristics and symptomology.

Stanford-Binet Intelligence Scale: Fifth Edition (SB-5) is a standardized test that measures intelligence and cognitive abilities in children and adults. For this study, an IQ score, used for eligibility, was determined from two subtests, yielding a nonverbal and verbal IQ score. The abbreviated IQ scores are highly correlated with full-scale IQ scores. The SB-5 has high reliability.

Primary outcome measure

The Friendship Survey. The Friendship Survey is a two-page questionnaire used to assess children's friendships and networks of peer relationships in the classroom (Cairns & Cairns, 1994). The survey was administered in the general education classroom during the school day and collected at three time points (a) before the start of intervention (about 2–3 weeks); (b) at the end of treatment (within 2 weeks); and (c) at follow-up (6–8 weeks after the intervention). Only children with ASD and their typically developing classmates who gave written assent and whose parents gave written consent completed the survey. Surveys were given to the class only after at least half of the class provided consent and assent. Surveys completed by half of the class (reporting on all members of the class) are required to be a valid measure of classroom social

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networks (Cairns & Cairns, 1994). Survey information from the first data collection period was used in conjunction with teacher nomination to select peer mentors for the ENGAGE groups.

Graduate student researchers facilitated the administration of the survey. They introduced the surveys to the children by telling them that we were interested in children's friendships at school, and that the information was confidential. We asked them to assign both girls and boys to groups who play together at recess, and to be sure to include themselves in the groups in which they belong. The researchers were available to answer questions or to help students that had difficulty reading or filling out the survey independently.

Children were asked to list names of children who were their friends. Then they were asked to list all the names of children in their class *who hang out together*; this free-recall method is used rather than asking respondents to place each peer's name in a group, because the relative strength of students' tendency to recall classmates is an important indicator of the peers' prominence in the social structure. Following Cairns' (1994) procedure, we derived profile scores for each child. Profile scores indicate the number of times that a child was listed as belonging to a group. The Social Network Salience (SNS) score refers to a ratio score that describes a child's social prominence within his or her own classroom across multiple time points (Kasari et al., 2011). To measure the change in children's SNS as a result of intervention, we divided the child's profile score by the highest profile score within that child's class at entry. This method was more suitable to measure change because the range of individual centrality scores varied by classroom. Thus, classroom differences were normalized using the SNS ratio scores.

Secondary measures

Playground observations. Blinded observers conducted the behavioral observations at baseline, during intervention—at weeks 3 and 6 (year 1) or at 4 weeks (year 2), postintervention, and follow-up for a total of five (year 1) or four (year 2) observations. We used the Playground Observation of Peer Engagement (POPE), an interval coding system that identifies durations of joint engagement with peers (e.g., having a conversation, playing a game) or solitary play (isolated from peers) (Kasari, Rotheram-Fuller, & Locke, 2005). Coders were trained to high inter-rater reliability (>.80) prior to beginning the study using standard videotapes of peer interactions on the playground, and practice children at each site. Coder drift was assessed by coordinators at each site, and overseen by the study coordinator. Inter-rater reliability of 14 coders ranged from .76 to 1.00, average .87.

Nominations to and from child with ASD. From the Friendship Survey, the number of children nominated as friends by the child with ASD, as well as the number of classmates nominating the child with ASD as a friend were collected.

Potential moderator measures

Student Teacher Relationship Scale (STRS) (Pianta, 1992) is a standardized and valid instrument used to measure the teacher's perception of his or her relationship with a particular student in terms of closeness, dependency, and conflict. The STRS is a 28-item survey with item scores ranging from 1 to 5 (1 = definitely does not apply; 2 = not really; 3 = neutral, not sure; 4 = applies somewhat; 5 = definitely applies). Total subscale scores were used in analyses.

The Social Skills Improvement System (SSIS; Gresham & Elliott, 2008) is a standardized instrument measuring social skills, problem behaviors, and academic competence. This study used the teacher rating scale of problem behaviors subscale (30 items scored from 0–3, and yielding a standard score).

Statistical analysis

Student *t*-tests, wilcoxon-tests, chi-square tests, and Fisher exact tests were used for baseline comparisons between treatment groups (to assess the success of randomization). Generalized linear mixed models (GLMM) with main effects of treatment (ENGAGE and SKILL) and time, treatment by time interactions and subject level random intercepts were used to model the longitudinal trajectories of the outcomes, employing an identity link for continuous outcome variables and a log link function for count outcomes. Separate models were fit for each longitudinal outcome. Time was modeled continuously in phases where each phase consisted of 8 weeks. All available observations from each subject were utilized in modeling via the GLMM. All primary outcome models controlled for IQ (SB5) and the main effect of site. Treatment effect was defined as a significant interaction effect between the treatment groups and time during treatment (entry to end of treatment) and maintenance of treatment effect was defined as a significant improvement from entry to the 8-week follow-up only if a treatment effect was observed. In addition, maintenance effects were only evaluated as a secondary exploratory analysis as many of the children did not participate in follow-up. Interactions between site by time, and site by time by treatment allocation were also evaluated in post hoc analyses for the primary outcome models. All randomized children were included in all analyses in accordance with intention-to-treat principles.

The moderation effects of treatment with teacher and child relationships were also examined. Teacher's report of problem behaviors was measured using the SSIS and the relationship (i.e., closeness, dependency, and conflict) between teacher and target child was measured using the STRS.

Results

Preliminary analyses

Table 1 presents the demographic information of the target children along with the outcome measurements at entry. There were no significant differences between the treatment groups in terms of gender, age, overall IQ, and ADOS severity. There were significant differences in the distribution of race/ethnicity and language heard at home ($p < .001$ and $p = .006$ respectively) among the sites, expected given the geographic differences. Consequently, site main effect was controlled for in all models to adjust for the potential differences among sites.

Primary outcome

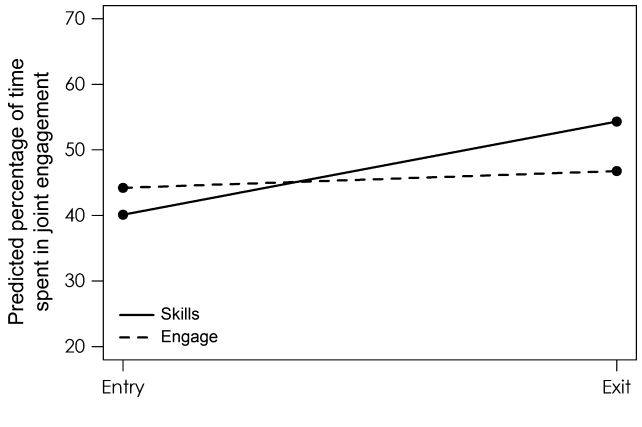
Social network salience. The primary outcome measure was the SNS ratio for the target child within his/her classroom. On average, there was no significant change in network salience over time by treatment group ($F(1, 109) = 0.01$, $p = .974$), however, there was a modest effect of overall improvements for both groups combined ($F(1, 109) = 3.64$, $p = .059$). There were no site by treatment by time interactions indicating that the effect of treatment did not vary by site ($F(3, 103) = 3.81$, $p = .149$).

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In addition, there was a moderated effect of teacher–child relationship after adjusting for the number of peer nominations to the child with ASD. There was a significant moderation of treatment effect with the STRS closeness score and conflict score at entry ($F(1, 96) = 13.55$, $p < .001$ and $F(1, 95) = 9.08$, $p = .003$). Children who were in the SKILLS group and whose teachers reported less closeness to them or reported higher conflict exhibited significant improvements in their SNS from entry to exit. Children whose teachers reported higher closeness scores to them and who were in the ENGAGE group experienced significant improvement in their network salience from entry to exit.

Secondary outcomes

Playground engagement. The percentage of time spent in joint engagement with peers was evaluated using a linear mixed model. On average, all the children significantly increased in their percentage of time spent in engagement with their peers from the start of treatment to the end of the treatment ($F(1, 121) = 9.03$, $p = .003$). In addition, there was a significant treatment effect where the children in the SKILLS group improved significantly more than the children in the ENGAGE group ($F(1, 121) = 4.32$, $p = .040$) (see Figure 2). Similarly, there was a significant treatment effect on isolation during recess ($F(1, 121) = 10.01$, $p = .002$) where children in the SKILLS group significantly decreased in the percentage of time spent in isolation compared to the children in the ENGAGE group. No site effects (interaction between site with time, and site by time by treatment interactions) were found for the percentage of time spent in isolation, but site effects were found for the percentage of time spent in engagement ($F(3, 115) = 5.18$, $p = .002$). Site 1, site 2, and site 4 had similar patterns where children in the SKILLS group improved more in time in engagement as compared to the children in the ENGAGE group. Children in site 3 showed a different pattern where children in the ENGAGE group showed greater improvements as compared to the children in the SKILLS group. Further exploration revealed that only six children in the SKILLS group remained in the study until the end of the treatment in site 3; thus, these results should be interpreted with caution.



Figures 2 Predicted time in joint engagement across group and time

Both the teacher report of SSIS problem behaviors ($F(1, 95) = 4.35$, $p = .04$) and STRS closeness ($F(1, 107) = 5.85$, $p = .017$) at entry moderated observed peer engagement. Children with high scores on behavior problems and low levels of closeness made greater gains in observed peer engagement if they were in the SKILLS group.

The changes in engagement were not maintained at the follow up for the SKILLS group. Nearly half of the sample was lost given the school year calendar, but for those who remained, follow-up engagement was not significantly greater than entry engagement; ($F(1, 79) = 2.80$; $p = .098$).

Number of children nominated by child with ASD. The number of children nominated by the child with ASD did not change with treatment ($F(1, 112) = 1.99$, $p = .161$). No site effects (interaction between site with time, and site by time by treatment interactions) were found.

Number of children who nominated the child with ASD. The number of children who nominated the child with ASD also did not change significantly with treatment ($F(1, 110) = 1.52$, $p = .220$) and there were no differences by site.

Discussion

This study compared social skills interventions in the school setting for elementary-aged children with ASD. Contrary to expectations, results yielded more consistent support for a skills-based, social skills group consisting of all children with social challenges than a mixed (typical and ASD) social skills group. There were no main effects of treatments on the primary outcome of peer-rated social networks but there were significant moderators. Social networks were significantly improved for children in the SKILLS group who entered the study with high conflict and low closeness with their teachers, or if children received high teacher closeness scores and they received the ENGAGE group. These may be among the first data suggesting that children with different quality teacher–child relationships benefit more from specific types of social skills interventions.

Main effects of treatment were found for two secondary outcomes. Children in the SKILLS group demonstrated significantly more engagement with their peers and less solitary play compared to the children in the ENGAGE groups as rated by blinded observers. Similar to the social network data, children who began treatment with low teacher closeness and high problem behaviors benefitted more from the SKILLS intervention. These data add fur-

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ther support for the push toward personalizing social skills interventions in real world contexts for maximum benefits.

Contrary to our expectations, then, these data suggest that an adult-led didactic social skills approach with children from different grades and classes was more effective for improving peer acceptance and engagement on the school playground than a naturalistic, activity-based intervention for children with and without ASD from the same classroom. While these same types of didactic, adult-led groups in clinic settings have rarely generalized to the school setting, our findings indicate that when conducted at school, they appear more effective than inclusive, interest-based social skills groups for children with ASD and their classmates. These findings are particularly impressive because the children in the SKILLS groups were from different grades and different classrooms. SKILLS children may not have been on the playground at the same time, nor were they able to nominate children on the social network measure from their SKILLS groups as they were not members of the same classroom.

There are several reasons why SKILLS groups may be more effective than the ENGAGE groups. One is the SKILLS group provided direct instruction in social skills that were immediately practiced within the group, an approach that has proven very effective with children with ASD in the clinic (Laugeson et al., 2012). Because the groups were conducted in the school setting, children may have gained more confidence to execute their social skills with children on the playground regardless of whether they were the same children they had practiced with in the group. Another reason is that the SKILLS group brought together children who were likely more similar to each other (homophily), and in a group format that allowed children to get to know each other who may not have previously (propinquity). Thus, homophily and propinquity criteria for developing peer relationships may have been more highly met in the SKILLS group than the ENGAGE groups. However, to truly understand the influences of homophily and propinquity on children's peer relationships, it will be important for future research to include interviews with the children themselves.

Findings from this study differ from previous peer-mediated interventions carried out in schools. In the Kasari et al. (2012) study, an 'indirect' peer-mediated intervention approach resulted in better child engagement outcomes compared to an adult–child one-on-one social skills intervention. The peer-mediated intervention was indirect because the child with ASD was not directly integrated with the typical peers; rather the typical peers were charged with engaging any child on the playground who was having social difficulties. This study expanded on the previous study by combining children with ASD directly with their peers in shared activities, and common interests. This approach was expected to improve the social engagement of the children with ASD as a result of these direct experiences with some precedent for this approach from single case designs (e.g., Koegel, Vernon, Koegel, Koegel, & Paullin, 2012). However, these groups likely need to be carefully considered as qualitative data of one all-girl ENGAGE group highlighted some negative ramifications of group interactions. In this group, the girl with ASD became more isolated and excluded due to her topic perseveration that became increasingly annoying to her classmates. The exclusion was very subtle and only apparent by careful, micro coding in a qualitative study (Dean, Adams, & Kasari, 2013). The mixed-group format may have inadvertently strengthened the connections between the typical classmates, and further highlighted the differences of the girl with ASD, serving to further separate her from the others. Potential negative side effects of behavioral interventions may be an important topic for future research.

Although this study is one of the larger social skills interventions, there are limitations. Despite comparing treatment conditions, we could not disentangle the active ingredients of either treatment. A future study should employ a more traditional 2×2 treatment design to better understand the contributions of the teaching approach and group composition. Another issue is that close to 10% of the sample dropped from the study during the treatment phase due to unforeseeable circumstances leading to significant protocol deviations. Deviations mostly reflect uncontrollable events such as teachers changing classrooms in the middle of the school year, children moving away, and children missing too many intervention sessions. Another limitation is the significant attrition by follow-up where only 50% of the children remained. These numbers are due largely to the ending of the school year, and subsequent changes in classroom to the next school year. A more cost-effective and real-world approach would be to train school staff to implement the interventions for potentially greater benefit. Overall, these data speak to the difficulty of conducting studies in the school setting and the significant resources required to obtain meaningful data (Kasari & Smith, 2013).

In summary, inclusion offers children with ASD the potential of interactions with typical peers and opportunities for participation in ordinary childhood experiences such as playing with others at recess. However, we have less information on how to support these opportunities. This study applies two different approaches to this challenge and shows that direct instruction in a supportive homogeneous group of children is more likely to generalize to the school playground than indirect, activity-based activities among mixed groups of children from the same classrooms.

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Key points

- The study employed randomized clinical trial comparing two methods for delivering social skills interventions in schools with a large, diverse sample of children with ASD (aged 6–11 years).
- There was a differential treatment effect on observable social behavior during recess. The SKILLS intervention (didactic group composed of all children with ASD from different classes and grades) was more effective than the ENGAGE intervention (interest- based groups composed of typical children and children with ASD from the same classroom).
- The teacher–child relationship moderated the effect of treatment on social network connections. Children with low teacher child closeness or high conflict improved more in their social connections if they received the SKILLS intervention, whereas children with higher teacher–child closeness improved more if they received the ENGAGE intervention.
- Data from this trial highlight the importance of individualizing treatment approach based on child characteristics.

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